

Curvilinear position

Glasgow airport 05/23 length.

i) 05: $55^{\circ}51'48.33''$ N, $4^{\circ}26'56.59''$ W

ii) 23: $55^{\circ}52'41.7''$ N, $4^{\circ}25'18.42''$ W

$$\Rightarrow L_{05} = 55.86348^{\circ}, L_{23} = 55.87825^{\circ}$$

$$\lambda_{05} = -4.44905^{\circ}$$

$$\lambda_{23} = -4.42178^{\circ}$$

Assume $h_0 = 0$. The transverse radius of curvature for each end are:

$$R_E(L_{05}) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_{05}}} = \frac{6378.137}{\sqrt{1 - e^2 \sin^2(55.86348^{\circ})}} = 6,392,813 \text{ m}$$

$$R_E(L_{23}) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_{23}}} = \frac{6,378,137}{\sqrt{1 - e^2 \sin^2(55.87825^{\circ})}} = 6,392,818 \text{ m}$$

Then:

$$\begin{aligned} x_{e05}^e &= R_E(L_{05}) \cdot \cos L_{05} \cdot \cos \lambda_{05} \\ &= 6,392,813 \cdot \cos(55.86348^{\circ}) \cdot \cos(-4.44905^{\circ}) \\ &= 3,576,623 \text{ m} \end{aligned}$$

$$\begin{aligned} y_{e05}^e &= R_E(L_{05}) \cos L_{05} \sin \lambda_{05} \\ &= -278,286.4 \text{ m} \end{aligned}$$

$$\begin{aligned} z_{e05}^e &= (1 - e^2) \cdot R_E(L_{05}) \cdot \sin L_{05} \\ &= 5,255,927 \text{ m} \end{aligned}$$

$$x_{e23}^e = R_E(L_{23}) \cdot \cos L_{23} \cdot \cos \lambda_{23}$$

$$= 3,515,398 \text{ m}$$

$$y_{e23}^e = R_e(L_{23}) \cdot \cos L_{23} \cdot \sin \lambda_{23}$$

$$= -276,147 \text{ m}$$

$$z_{e23}^e = (1 - e^2) \cdot R_e(L_{23}) \cdot \sin L_{23}$$

$$= 5,253,116.5 \text{ m}$$

$$\|r_{eos}^e - r_{e3}^e\| = 355 \text{ m}$$